2.3 Startup Operation

2.3.1. Startup Interlock

The DECSS system checks the items listed below as the conditions for the engine to start up safely. If any one of the conditions is not satisfied, you cannot start the engine.

(a) The turning safety switch is on.
- The turning safety bar is set in place as prescribed.
- The turning motor does not mesh with the gear.

(b) The handle switch is set to the RUN position (only if the operation position is REMOTE). (When the operation position is LOCAL, the position of the handle switch does not matter.)

(c) The engine is not operating at all.

(d) No startup command has been issued.

(e) No stop command has been input.
- No local STOP is input.
- No remote STOP is input.

(f) No START FAILURE has occurred.

(g) No emergency stop (TRIP) has occurred.

(h) The 2 systems of the engine rotational speed pickup sensors are working for detection. (The lubricating oil pressure (priming pressure) is 0.01 MPa or larger. (However, you can start the engine if 24 hours has not elapsed since the pressure becomes lower than 0.01 MPa.) If the lubricating oil pressure sensor is abnormal, condition (h) is excluded from the conditions.

If the startup conditions are satisfied, the "READY TO START" LED indicator on the engine condition display turns on to indicate that you can start the engine. If a LCD gauge board is in the system, the READY TO START item in the information window shows "READY" on the display.

NOTE)
During a power outage, the engine stops. As the power source is not available, it is impossible to operate the priming pump. Therefore, the lubricating oil pressure becomes 0 MPa. This should be included in the conditions of the startup interlock. However, in this system, item (h) is excluded from the conditions of the startup interlock to enable the startup if 24 hours has not elapsed since the lubricating oil pressure becomes lower. If priming the lubricating oil is not conducted for 24 hours or longer, for example, in the case of a startup from a dead ship, the startup conditions are not satisfied. So, the startup is not possible. Make sure to conduct priming

before starting the engine in order to sufficiently lubricate each bearing of the engine. Then, start the engine.

2.3.2. Starting the Engine from the Engine Side (LOCAL Position)

When you have switched the ENGINE MODE switching switch on the engine condition display to the LOCAL position, you can start the engine by following the steps listed below:

① Execute the priming of the lubricating oil.
② Open the valves of the startup air and the control air.
③ Set the operation lever to the RUN position.
④ Keep on pushing the START push button switch. The starting solenoid valve and the F.O. control solenoid valve work in the prescribed sequence. Thus, the engine startup operation starts.
⑤ The engine rotational speed increases. When the speed reaches the low speed rotational speed (setting—S#14:300min^{-1}), then the starting solenoid valve turns off. (The solenoid may turn off at 150 min^{-1} depending on the setting.)
⑥ When the low speed is reached, the F.O. control solenoid valve turns off in a while. According to the control by the governor, the rotational speed of the engine is stabilized to the rated rotational speed.

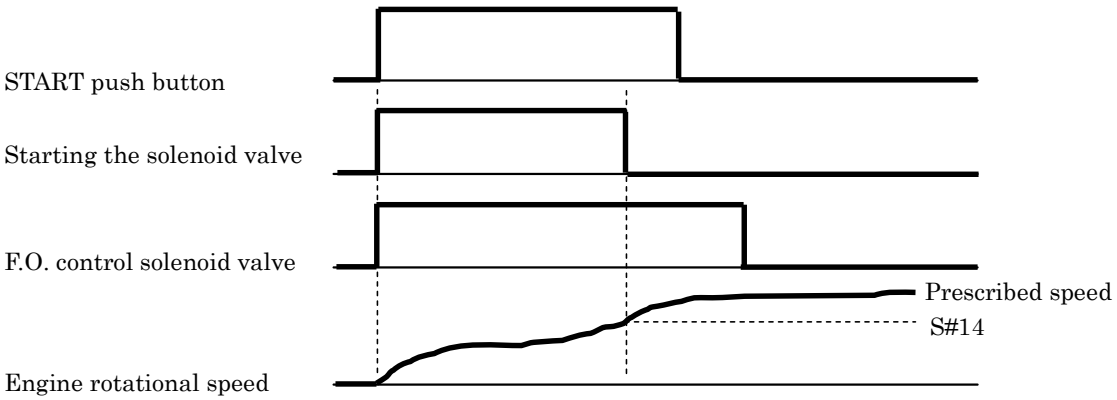


Fig. 38 The time chart of the startup sequence

2.3.3. Air Run

If you have switched the ENGINE MODE switching switch on the engine condition display to LOCAL, you can conduct an air run by following the steps listed below. You can conduct an air run to clear the unburnt gas in the cylinder.

① Execute the priming of the lubricating oil.
② Set the operation lever to the STOP position.

③ Set the pressure indicator to FULL OPEN.

④ Open the valves of the startup air and the control air.

⑤ Keep pushing the START push button switch on the operation panel for 3 to 4 seconds. The starting solenoid valve and the F.O. control solenoid valve start operation in the prescribed sequence. The engine starts a startup operation. The startup air turns the engine. However, the fuel is not injected. So, the rotational speed of the engine does not increase.

⑥ Release the START switch. The starting solenoid valve turns off immediately. The F.O. control solenoid valve, on the other hand, turns off its solenoid valve after the engine stops completely.

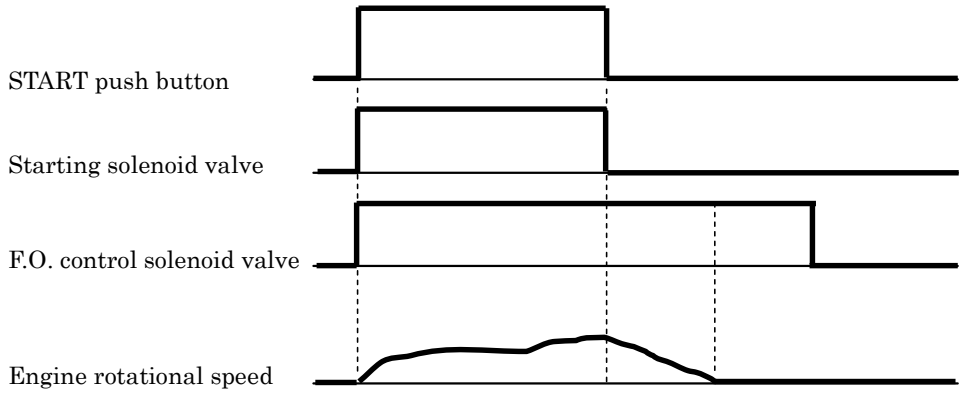


Fig. 39. The sequence time chart of the air run

2.3.4. Starting the Engine from the REMOTE Position

If you have switched the ENGINE MODE switching switch on the engine condition display to REMOTE, you can start the engine by following the steps listed below.

① Execute the priming of the lubricating oil. (Select the AUTO mode.)

② Open the valves of the startup air and the control air.

③ Set the operation lever to the RUN position.

④ Turn on (close) the Remote Start signal input. The starting solenoid valve and the F.O. control solenoid valve start operation. The engine starts operation.

⑤ The engine rotational speed increases. When the speed reaches the low speed rotational speed (setting—S#14:300min^{-1}), then the starting solenoid valve turns off. (The solenoid may turn off at 150 min^{-1} depending on the setting.)

⑥ When the low speed is reached, the F.O. control solenoid valve turns off in a while. According to the control by the governor, the rotational speed of the engine is stabilized to the rated rotational speed.

2.3.5. Startup Failure

If the rotational speed does not increase normally when the engine is starting up, then a startup failure is detected and the engine starts its stop operation.

When the REMOTE mode is active at the time of a startup, if the engine rotational speed does not reach or exceed the specified low speed (S#14:300min-1) before the startup failure detection timer expires, then the engine starts its stop operation and outputs the Start Failure alarm. The "START FAILURE" LED indicator lamp on the engine condition display turns on. If the LCD gauge board is available, the message appears on the information window. After a startup failure is detected, the emergency stop solenoid valve and the shutdown solenoid valve turn on automatically. The engine starts its stop operation. The engine stops and remains in the status of a startup failure.

Find the cause of the startup glitch. Remove the cause. To reset the startup glitch, set the operation lever (HANDLE SW) to the OFF position. The status of the startup glitch is reset. You cannot restart the engine until you reset the startup glitch.

NOTE)
In the LOCAL mode, the startup failure is not detected. If an error is found while the engine is starting up from the operation on the engine side, and if it is necessary to stop the engine, then use the STOP button or the EMERGENCY STOP button on the engine condition display or set the operation lever (HANDLE SWITCH) to the stop position to stop the engine.

<u>If a startup failure occurs, make sure to follow the instructions described on the manual and take corrective actions.</u>

2.4 Operation

When the engine starts up normally, and if the engine is controlled to the prescribed speed by the governor, then "ENGINE RUN" LED turns on in green on the engine condition display. When the LCD gauge board is available, the message ENGINE RUNNING is displayed on the information window. In addition, the operation signal is output to the contact with the external device.

In the DECSS system, the operation time is counted up and accumulated while the engine is in the status of operation. You can push the up arrow key on the engine condition display to view the total operation time. (The MODE switch needs to be set to DISPLAY MODE. The selection number is "01.")

The "ENGINE RUN" LED on the engine condition display turns on when the reading is 300 min^{-1} or higher. After a startup signal is input, and before 300 min^{-1} is reached, the system is in the status of starting up. After a stop signal is input, and before the engine stops completely, the system is in the status of stopping. In both of the status, the LED blinks.

2.5 Stopping Operation

When you push the STOP push button switch on the engine condition display, or a Remote Stop signal (close) is accepted, the engine stops by following the steps listed below:

① The shutdown solenoid valve turns on. The fuel cut device works to stop the fuel supply to each cylinder.

② After the engine stops completely, the shutdown solenoid valve turns off. The operation of the fuel cut device is reset to the original status. When, however, the STOP push button on the operation panel is being operated, the shutdown solenoid valve continues to be on.

③ The operation lever (HANDLE SWITCH) is shifted to the STOP position.

2.6 Emergency Stop Operation

2.6.1. Manual Critical Stop (Manual Shutdown)

If you push the EMERGENCY STOP push button on the engine condition display, or if a remote emergency stop signal is input from an external device, then the emergency stop solenoid valve turns on. The fuel cut device operates to cut the fuel supply, and the engine stops (emergency stop). At the same time with this, the Emergency Stop alarm is output to the external device, and the "EMERGENCY STOP" LED on the engine condition display turns on in red. If the LCD gauge board is available, the message is displayed in the information window. When the emergency stop solenoid valve turns on, the shutdown solenoid valve turns on, too.
To reset this status, the engine rotational speed should be naught completely. Set the operation lever (HANDLE SWITCH) to the OFF position. Then, the status of the emergency stop is cancelled. The emergency stop solenoid valve turns off. The normal status is recovered. You cannot restart the engine until the operation described above is completed.

2.6.2. Automatic Critical Stop (Automatic Shutdown)

If any failure occurs to the engine, and if any of the conditions listed below is met, then the DECSS system activates the safety function. The emergency stop solenoid valve turns on automatically to cut the fuel supply for an emergency stop. At the same time with this, a serious-failure signal is output to the external device, and the LED indicator on the engine condition display turns on in red. If the LCD gauge board is available, a message is displayed on the information window.

Table 8 Emergency shutdown

No.	Item	Monitoring
1	***OVER SPEED stop*** (The engine speed is excessive.)	Always
2	***H.T. WATER TEMP. HIGH stop*** (The temperature of the high-temperature cooling water is abnormally high.)	During engine operation (From #14T:300min^{-1}+TIMER to a stop operation)
3	***LUB. OIL PRESS. LOW stop*** (An abnormally low lubricating oil pressure is detected.)	During engine operation (From #14T:300min^{-1}+TIMER to a stop operation)
4	***OIL MIST DENSITY HIGH stop*** (The oil mist detector detects a high concentration alarm (high-alarm).)	Always

When the emergency stop solenoid valve is on, the shutdown solenoid valve turns on, too. Find the cause of the emergency stop, and remove the cause.

If an emergency stop occurs, make sure to follow the instructions described on the manual and take corrective actions.

To reset this status, the engine rotational speed should be naught completely. Set the operation lever (HANDLE SWITCH) to the OFF position, Then, the status of the emergency stop is cancelled. The emergency stop solenoid valve turns off. The normal status is recovered. You cannot restart the engine until the operation described above is completed.

3 Maintenance and Inspection

Before you replace a sensor, a cable, and/or a component in the DECSS system, turn off the power to the control unit. Before you replace the main part of the control unit, turn off the source of the power supply.



DANGER

・Do not turn off the power switch of DECSS while the engine is running.
If you make an incorrect operation or incorrect setting, not only does the engine become unable to work properly but also a serious accident may result.



CAUTION

・Before you replace a sensor and/or a cable, make sure to turn off the power switch of DECSS.
・Perform your maintenance and inspection work only when the engine is not running.
・If any abnormality is found during an inspection, see the next section to take necessary corrective actions.

1) Inspection for Errors (Conduct the inspection whenever possible.)

(1) Check if there is any loose connection and/or any loose bolt and/or any damage in and/or on the control unit, the junction box that houses the control unit, the condition display, and the LCD gauge board.

(2) In addition, check that there is not abnormal heat emission, abnormal vibration, and/or abnormal sound.

2) Inspection of the Display (Conduct the inspection whenever possible.)

(1) Push the lamp test button on the engine condition display to see if all the LEDs turn on.

(2) If the LCD gauge board is available according to the specifications, check that the LCD display works correctly. In particular, check if there are any pixels missing, check if there is any unstable screen image, and check if any backlight that does not turn on.

(3) Check the system fail items on the engine condition display.

(4) Check that all the trip items on the engine condition display is not lit.

(5) If the LCD gauge board is available according to the specifications, check that the information window is not showing any system error or any trip item.

3) Inspection for Loose Screws (Conduct the inspection every 6 months.)

(1) Inspect the wire isolator (anti-vibration device) to see if there is any interference and/or breakage.

(2) Inspect the screws that fasten the lid on the board to see if any of them is loose.

(3) Check the cable connectors to see if any one is loose or disconnected.

4) Inspection of the Operations of the Functions (Conduct the inspection every 6 months.)

(1) While the engine is not running, push the stop button to check that the fuel cut solenoid valve works and the fuel cut device operates.

(2) While the engine is not running, push the emergency stop button to check that the emergency stop solenoid valve and the fuel cut solenoid valve operate, and check that the fuel cut device works.

(3) Use the air run operation to check that the startup solenoid valve and the fuel control solenoid valve work.

5) Miscellaneous Checks and Maintenance Works

(1) It is recommended that you should ask for maintenance inspections every 4 years to ensure the reliability of each component included in the DECSS system. Also, it is recommended that the replacement should be made every 8 years.

(2) If you are going to store the system or if a construction period continues, cover the entire system with some plastic cover or the like for protection against water.

(3) While an instruction is ongoing for example, mount a panel to protect the board and mount a protection bracket to prevent any breakage.

(4) During your maintenance work for example, do not step on the LCD gauge board. Doing so may deform the wire isolator (used to prevent vibration) very much, which may obstruct the anti-vibration performance.